Autonomous Vehicle

A Study on Analyzing the Security and Privacy Implications of Existing and Emerging Sensors

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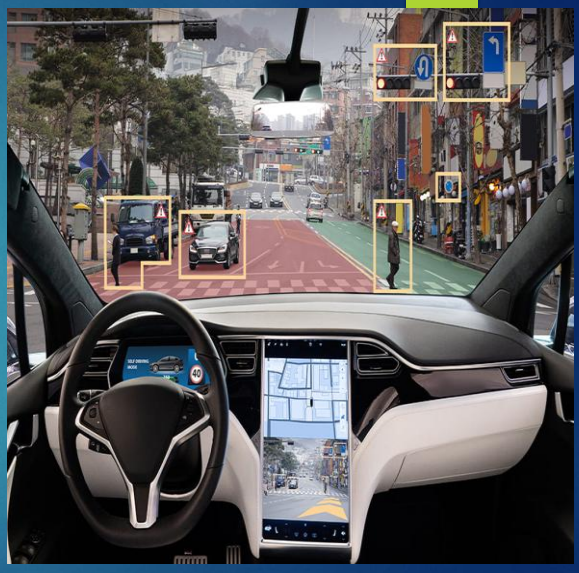

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INTRODUCTION :

- An autonomous vehicle (AV) is a vehicle that can navigate and operate without human intervention. AVs, also known as self-driving cars, use sensors, cameras, radar, and artificial intelligence (AI) to perceive their environment and make decisions.
- **GNSS** (like GPS) – to know location.
- Self-driving cars are able to move independently thanks to special software and sensors. The software controls the operation of all car systems: turning the steering wheel, changing gears, gas and brakes.
- Sensors collect information about the environment, which forms the basis for the actions of the car.

Autonomous vehicles are classified into six levels (0-5) by the Society of Automotive Engineers (SAE):

Level 0: No automation (fully human-driven).

Level 1: Driver assistance (e.g., adaptive cruise control).

Level 2: Partial automation (driver still required, but car can steer and accelerate).

Level 3: Conditional automation (car can drive itself in some conditions, but driver must be ready to take over).

Level 4: High automation (car can operate without human intervention in most conditions).

Level 5: Full automation (no human driver needed at all).

Security and Privacy in IoVs

Understand the Problem – Wireless Data Is Vulnerable

- Hackers could: Steal data (e.g., location, direction, speed)
- Send fake signals
- Control sensors remotely

- Solutions to Make Autonomous Vehicles Safer

Area	Solution
🔐 Data Security	Encryption, Blockchain, Authentication
🧠 Privacy	Federated Learning, Differential Privacy
🚨 Threat Detection	AI/ML, Anomaly Detection
🚗 Real-time Safety	Cloud + Fog, Redundant Sensors
🌐 Future-proofing	Post-Quantum Cryptography

- Use Strong Encryption

  Encrypt all communication between sensors, vehicles, and servers
- Use AI and Machine Learning for Threat Detection

  Can detect unusual activity (e.g., a fake signal or hacking attempt).
- **Federated Learning to Protect Privacy**

  Vehicles train AI models **locally** using their own data.
- Use Cloud and Fog Computing for Fast, Secure Processing

  Fog computing processes data closer to the vehicle (faster).

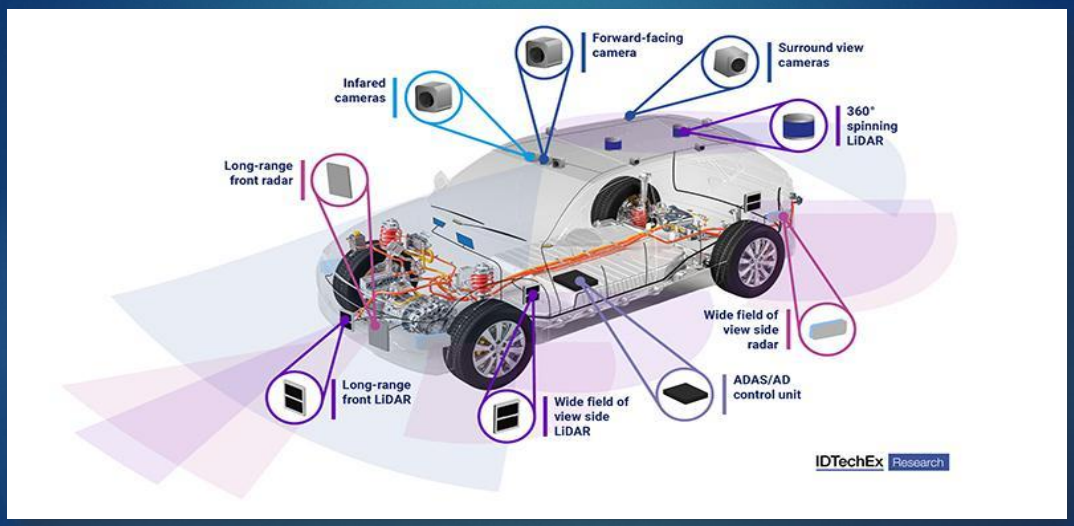
  Cloud computing provides storage and large-scale analysis.
- **Access Control & Authentication Systems**

  Only verified users, sensors, and vehicles can access the system
- **Anomaly Detection Systems**

  Constantly monitor for odd behavior in vehicle systems.
- Redundant and Secure Sensor Networks

  Use **multiple types of sensors** (LiDAR + RADAR + GPS).

AUTONOMOUS VEHICLE PARTS :



- Technological Advancements
- Sharper perception and decision-making: AI algorithms are more adept at understanding environments with advanced sensors and robust machine learning.
- Faster, more autonomous operation: Edge computing enables on-board AI processing for quicker decisions and greater independence. Enhanced safety and reliability: Redundant systems and rigorous fail-safe mechanisms prioritize safety above all else
- Surging demand for AI expertise: Specialized courses and degrees in autonomous vehicle technology will cater to a growing need for AI, robotics, and self-driving car professionals.

Pros and Cons of AV

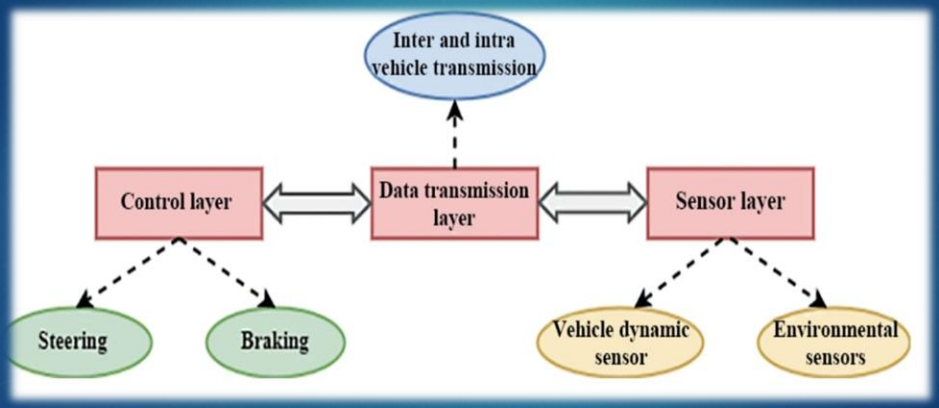
PROS

- Reduce motor vehicle fatalities
- Save people time
- Decrease traffic jams
- Prevents drunk driving accidents
- Reduced Parking Needs

CONS

- Wide adoption necessary to reduce traffic and accidents
- Job loss in the transportation industry
- Difficult to Determine fault in an accident
- High Initial costs
- Potential loss of jobs in transportation

Diagram of the three-tier automated EV structure.



Explanation of Block Diagram :

- Central core layers : Gathering information from the vehicle's surroundings and its own internal state
- Vehicle dynamic sensor: monitors the speed, acceleration, yaw rate, steering angle etc.
- Environment sensors: perceive the external environment, such as Cameras, Radar, Lidar, Ultrasonic sensors, GPS, etc.

- Data Transmission Layer: This acts as the central hub for communication. It receives data from "Sensor layer". It transmits data to "Control later".
- Steering : Controls the vehicle's direction
- Braking : Controls the vehicle's deceleration and stopping

Applications :

- 1.Transportation & Mobility
- 2.Agriculture
- 3.Industrial & Construction
- 4.Military &Defense
- 5.Healthcare & Emergency Services

Conclusion :

- The study highlights key challenges in automobile networks, particularly security, user privacy, and trust. It examines cybersecurity threats, supply chain risks, and issues in the Internet of Vehicles (IoVs). Findings from various datasets reveal that privacy concerns are highest in collaborative systems. The paper reviews different security threats and solutions, emphasizing their impact on the safety of autonomous electric vehicles in real-world environments. It stresses the importance of analyzing the system as a whole, not just individual components. Future research should focus on cyber-physical system architectures to address data confidentiality and communication security

Thank You

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